EIKON

Smart Attendance System

SETUP & DEPLOYMENT GUIDE

Component	Technology
Desktop Hub	PyQt6, Custom Title Bar, DWM Shadow
Face Recognition	InsightFace buffalo_sc, 512D Embeddings
Web Dashboard	Flask, SocketIO, Live Camera Stream
Kiosk Terminal	Fullscreen, Anti-Spoof, Auto-Capture
Mobile Admin	Flutter, REST API, Configurable Server
AI Assistant	Groq LLM (Nixa), Natural Language Queries
Database	MySQL 8.0, SQLAlchemy ORM

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Prerequisites

What you need before running setup:

Requirement	Check	Source
Python 3.10+	python --version	python.org/downloads
MySQL Server 8.0+	mysql --version	dev.mysql.com/downloads
Webcam	Built-in or USB	Any USB webcam
Internet (once)	First run only	AI model auto-downloads ~30MB

When installing Python, check "Add Python to PATH" on the first screen. Without this, setup.bat cannot locate Python.

One-Click Setup

The entire environment configures itself. No manual steps required.

1. Open the Eikon folder
2. Double-click **setup.bat**
3. Enter your MySQL root password when prompted
4. Choose camera source (0 = built-in, 1 = external USB)
5. Wait for all checks to pass (takes 3-8 minutes)

The script creates a virtual environment, installs all packages, generates the database schema, seeds default subjects, and verifies connectivity. You do not need to touch the terminal yourself.

Launching Eikon

After setup completes, activate the environment and launch:

```
venv\Scripts\activate

python run.py
```

This opens the **Eikon Hub**, the central command center. Everything else is managed from inside the Hub. You do not need to run any other scripts manually.

The Hub provides five action cards:

Card	What It Does
Register	Opens guided face capture kiosk. Enter student name and reg number, camera captures 30 images autom

Card	What It Does
Train	Generates 512-dimensional face encodings from captured images. Output streams live in the Developer C
Recognize	Launches the attendance scanning kiosk. Asks for subject ID, then opens the camera for live face matchin
Kiosk	Same as Recognize but launches directly in fullscreen mode for door-mounted tablets.
Dashboard	Starts the Flask web backend and opens the admin dashboard in your browser at localhost:5000.

The Hub also provides: camera source picker, dark/light theme toggle, demo data seeder, live backend status LEDs, session uptime, scan counter, and a developer console with color-coded logs.

First-Time Workflow

After setup.bat completes, follow this sequence inside the Hub:

01 Launch the Hub

Run python run.py to open Eikon Hub.

02 Register students

Click the Register card. A dialog asks for reg number and name. The kiosk opens in capture mode with guided head-turn instructions. Repeat for each student (minimum 2-3).

03 Train the model

Click the Train card. The system reads all captured images, extracts face embeddings, and saves encodings.pkl. Progress streams live in the console.

04 Start recognition

Click Recognize. Enter the subject ID (default: 1). The kiosk opens in attendance mode. Faces are matched in real-time.

05 Open the dashboard

Click Dashboard. The Flask backend starts automatically and your browser opens localhost:5000 with live attendance data, camera feed, and analytics.

Recognition Feedback

The kiosk uses color-coded fullscreen feedback:

Color	Meaning
Green	Face recognized. Attendance marked successfully.
Amber	Recognized but late (past the grace period).
Blue	Already marked for today. Duplicate prevention.
Red	Unknown face. Not registered in the system.
Dark Red	Spoof detected. Printed photo or phone screen rejected.

Flutter Mobile App

The admin mobile app connects to the Flask backend over WiFi:

1. Both devices must be on the same network
2. Start the backend from the Hub (click Dashboard)

3. In the app, tap the connection icon and enter your laptop IP
4. The app connects on port 5000

Find your IP: run ipconfig in a terminal. Look for IPv4 Address under your WiFi adapter.

Configuration Reference

All settings live in the .env file. Key tunables:

Setting	Default	Effect
CAMERA_SOURCE	0	0 = laptop webcam, 1 = USB cam
RECOGNITION_THRESHOLD	0.55	Lower = more lenient face matching
ANTI_SPOOF_THRESHOLD	0.45	Lower = less strict spoof rejection
LATE_GRACE_MINUTES	10	Minutes after class start to mark late
MODEL_NAME	buffalo_sc	buffalo_l for higher accuracy (heavier)
GROQ_MODEL	llama-3.3-70b	LLM model for the Nixa AI assistant

Architecture

Module	Purpose
gui/eikon_hub.py	Central Hub: PyQt6 desktop dashboard with action cards, status LEDs, console, theme engine
gui/kiosk.py	Fullscreen kiosk: registration capture mode + attendance scanning mode
core/register.py	Face capture pipeline with guided head-turn instructions
core/train.py	InsightFace embedding generator. Reads images, outputs encodings.pkl
core/recognize.py	Real-time face matching loop with anti-spoof and duplicate prevention
app.py	Flask application factory with blueprint registration
app/routes/	API, dashboard, students, reports, attendance, AI chat endpoints
db/schema.sql	MySQL schema: users, students, subjects, attendance tables
config.py	Central config loader. Reads all settings from .env
run.py	Unified launcher: Hub / Web / Kiosk modes via CLI flags

Troubleshooting

"Python not found"

Reinstall Python with "Add to PATH" checked. Or use the full path to python.exe.

"MySQL connection failed"

Open Services (Win+R, services.msc) and start MySQL80. Verify password in .env.

"Camera not opening"

Close Teams, Zoom, or any app using the camera. Try CAMERA_SOURCE=1 in .env.

"Everyone shows as Unknown"

Click Train in the Hub after registering. Lower RECOGNITION_THRESHOLD to 0.45 in .env. Check lighting.

"Anti-spoof rejects real faces"

Lower ANTI_SPOOF_THRESHOLD to 0.4 in .env. Ensure face fills 1/4 of the frame. Avoid extreme lighting.

"InsightFace model download fails"

Check internet. If firewalled, download from github.com/deepinsight/insightface/releases. Place in ~/.insightface/models/buffalo_sc/

Your face is your ID.

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